

Topology-Sensitive Validation of a Constraint-Preserving Generative Surrogate for Zero Degree Calorimeter Showers

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Abstract

We investigate the validation of a constraint-preserving generative surrogate for single-neutron Geant4 showers in a fixed 6,790-channel Zero Degree Calorimeter readout. The hierarchical decoder enforces nonnegative deposits, prescribed channel multiplicities, and closure to sampled energy budgets. We evaluate one checkpoint trained on a 26,624-event pilot using a repeatedly inspected development bank of 10,000 conditions at 50–250 GeV. The generated events satisfy the recorded decoder-level numerical checks. The inclusive mean deposited energy differs from the reference by 1.0%, and the mean active-layer and active-channel multiplicities differ by 0.5% and -1.8% , respectively. Topology-sensitive observables give a different assessment: the mean number of interior inactive layers increases from 2.10 to 6.05, and the mean number of weak components on the readout graph increases from 23.42 to 59.39. The apparent agreement in inclusive response also contains compensating ECAL and HCAL biases. For this checkpoint, decoder-level validity and percent-level agreement in selected inclusive observables therefore coexist with substantially more fragmented longitudinal and spatial shower support. The result motivates treating numerical admissibility, agreement in low-order detector observables, and shower morphology as separate requirements in the validation of learned detector surrogates.

1 Introduction

Detailed Monte Carlo transport underpins detector design and high-energy-physics analysis, but calorimeter shower simulation is computationally demanding. Learned generative models are consequently being studied as detector-surrogate models, including normalizing flows, diffusion models, point clouds, graph networks, and conditional flow matching [1–6]. Their usefulness depends on physics validation: a generated event must be numerically admissible, but it must also reproduce the detector-level distributions and correlations needed by an analysis.

Sparse calorimeter data make this distinction especially important. A decoder can enforce nonnegativity, a prescribed number of active channels, and closure to its own sampled energy budgets. These constraints establish internal consistency of the generated event; they do not determine how energy is distributed across detector subsystems, layers, or neighboring channels. Similarly, agreement in inclusive response or mean occupancy does not by itself establish agreement in shower morphology.

Existing calorimeter studies evaluate generative models with one-dimensional shower observables, multivariate classifiers, model size, and generation time. CaloChallenge provides a broad comparison of these diagnostics [7], CaloDREAM discusses discrepancies visible at different levels of representation [6], and detector-integration studies extend validation to reconstructed observables [8]. Related work on ALICE Zero Degree Calorimeters models optical-fiber photon-count images [9]; the target considered here is instead a raw deposited-energy vector on a different readout. Graph diffusion and point-cloud approaches provide precedents for irregular calorimeter geometries [4, 5, 10], while ParaFlow illustrates the sensitivity of a parameterized calorimeter surrogate to upstream material [11]. Differences in target representation, detector geometry, and evaluation protocol prevent a direct numerical ranking with the present study.

This paper examines one validation failure mode in a hierarchical surrogate for a fixed, irregular Zero Degree Calorimeter readout with exact zeros, ganged channels, and 65 longitudinal layers. The contribution is threefold. First, we document the constraints enforced by the decoder and distinguish them from agreement with the Geant4 reference. Second, we quantify inclusive response and occupancy observables for one pilot-trained checkpoint. Third, we show that topology-sensitive longitudinal and readout-graph observables reveal substantially more fragmented generated showers despite near-agreement in the mean occupancies. Specifically, the active-layer and active-channel means differ by 0.5% and -1.8% , whereas interior inactive layers and weak graph components differ by factors of 2.88 and 2.54. These are descriptive comparisons on a development bank. They establish the coexistence of decoder-level validity, close selected means, and discrepant shower support for this artifact; they do not identify a causal architectural mechanism or characterize expected performance across retraining.

2 Simulation target and detector representation

2.1 Conditional target and scope

For each event, the available condition is a stored incident-neutron four-vector $p^\mu = (E, p_x, p_y, p_z)$. The model input is

$$c = (\log(1 + K_{\text{inc}}/100 \text{ GeV}), u_x, u_y, u_z, \log(E/\text{GeV})), \quad \mathbf{u} = \mathbf{p}/\|\mathbf{p}\|, \quad (1)$$

with $K_{\text{inc}} = E - m_n$ and nonzero momentum in the reported energy domain. The implementation maps zero momentum to zero direction by flooring the normalization denominator; the five features enter a learned condition encoder. The reference production records a fixed generator vertex, but the released metadata does not establish whether p^μ is a generator-level or calorimeter-entry state and does not fully specify upstream material. The operational target is therefore

$$p(\mathbf{Y} \mid p^\mu, \mathcal{S}), \quad (2)$$

where $\mathcal{S}$ denotes the fixed but incompletely documented reference-simulation configuration. If the four-vector is generator truth, this target implicitly includes stochastic upstream transport. The manuscript makes no claim about a different geometry, material configuration, particle species, or detector-entry state.

The preparation audit records 764,940 events spanning 0–300 GeV in incident kinetic energy, with deterministic event-hash partitions of 612,482 training, 76,158 validation, and 76,300 nominal test events. The reported checkpoint used a smaller pilot: 26,624 training events and 6,656 validation events for the training objective. A later training audit binds the 26,624-event population to the exact frozen configuration used here. The pilot contains 4.35% of the canonical training partition. The larger training roles defined for subsequent models do not describe this checkpoint.

All shower-comparison results use a separate 10,000-event bank drawn from the canonical validation partition with $50 \leq K_{\text{inc}} \leq 250 \text{ GeV}$. We denote it D_{dev} because its results were repeatedly inspected during development. One Geant4 event and one generated event are available per condition. The bank is disjoint from pilot training; its overlap with pilot validation is not quantified in the released aggregates. Checkpoint-selection losses and shower diagnostics therefore have distinct sample definitions. No nominal test event is used in this manuscript. Earlier project work inspected 40,000 nominal test events and a further draw containing 200 test events with unresolved overlap; we make no untouched-test claim.

2.2 Readout representation

The target $\mathbf{Y} \in \mathbb{R}_{\geq 0}^{6790}$ contains raw, nondigitized deposits summed by valid readout ID, with sentinel IDs excluded and no positive energy threshold. Its sum is modeled readout energy, not all energy lost

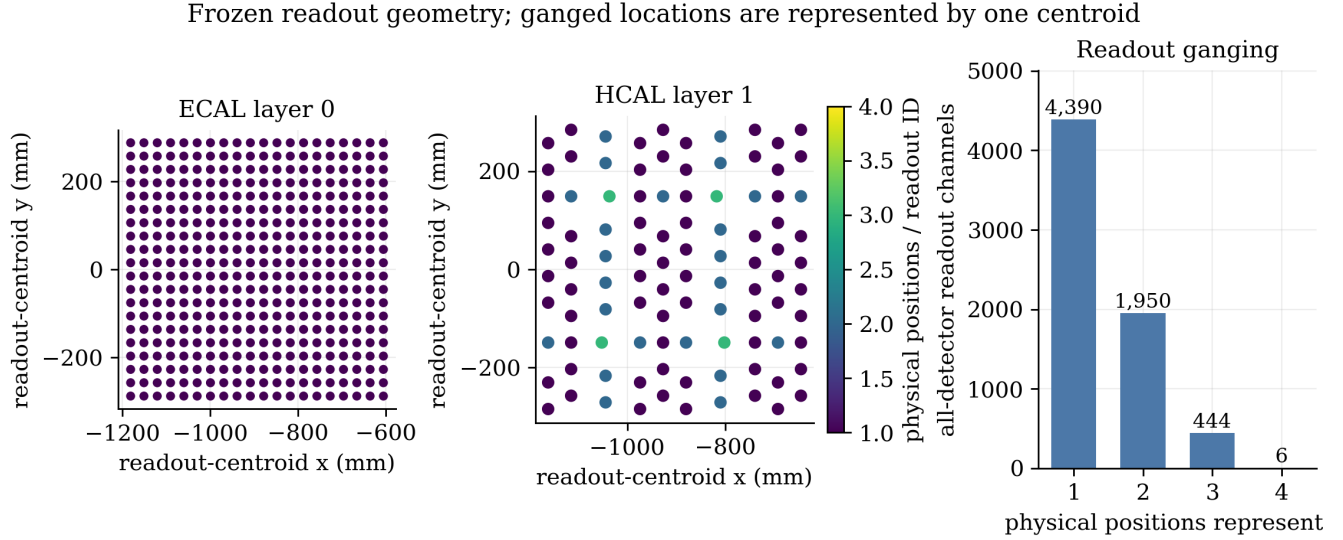


Figure 1: Frozen readout geometry. The left panels show ECAL layer 0 and HCAL layer 1. The ganging histogram is over all 6,790 readout channels; its first bar includes 400 ECAL and 3,990 HCAL channels.

by the incident neutron. Layer 0 contains 400 electromagnetic-calorimeter (ECAL) channels. Hadronic-calorimeter (HCAL) layers 1–63 contain 100 channels each and layer 64 contains 90, giving 6,390 HCAL channels. Across all 6,790 readout IDs, 4,390 represent one physical position, 1,950 represent two, 444 represent three, and six represent four. Thus 2,400 HCAL channels are ganged and 3,990 are not.

Each ganged ID is represented by the unweighted centroid of its physical positions. The model graph connects every channel to its eight nearest same-layer centroids. Adjacent layers receive four nearest-neighbor links per node in each direction, producing 107,920 directed edges. Both the model and the graph-connectivity diagnostic use this readout graph. Consequently, graph components are representation-specific and cannot resolve physical separation within a ganged channel.

3 Constraint-preserving generative surrogate

The 2,082,507-parameter generator follows the sequence in Fig. 2. It is probabilistically hierarchical but computationally dense over the readout graph. Training supplies reference upstream quantities to downstream component losses; evaluation samples every stage ancestrally. This teacher-forcing difference is part of the evaluated system.

3.1 Deposit and total-energy branch

A Bernoulli variable B first selects the nominal positive-deposit branch. Conditional on $B = 1$, a four-component Gaussian mixture models

$$z = \log(1 + T/10 \text{ GeV}). \quad (3)$$

The positive-reference training term is the mixture negative log density in z ; it is a latent-branch surrogate objective, not the likelihood of the final sampler. At generation time, a mixture draw z_s is transformed

3.3 Counts, graph support, and channel shares

A categorical head produces a count $K_\ell \in \{0, \dots, 400\}$ for every layer. Feasibility masks enforce $K_\ell = 0$ for inactive layers and $1 \leq K_\ell \leq N_\ell$ for active layers. A graph network scores every readout channel. Within each layer, Gumbel-Top- K_ℓ sampling selects exactly the requested number of channels [13]. The score temperature is fixed at one. Training combines a class-weighted support binary cross entropy with up to 256 randomly paired positive-negative ranking comparisons per event; this surrogate does not directly optimize the sampled top- k subset distribution.

The share flow acts on a dense 6,790-vector masked to the selected support. For nonempty support, a softmax converts terminal logits to shares. In real arithmetic the decoder is

$$Y_i = \begin{cases} D_\ell \frac{\exp q_i}{\sum_{j \in S_\ell} \exp q_j}, & i \in S_\ell, \\ 0, & i \notin S_\ell, \end{cases} \quad i \in \text{layer } \ell, \quad (9)$$

where D_ℓ is the sampled layer budget and S_ℓ the selected support. An empty support gives an all-zero layer without evaluating a 0/0 expression. For $K_\ell = 1$ the share is one. The implementation subtracts the largest selected logit before exponentiation; finite-precision underflow can still erase a selected deposit, so realized positive support is checked explicitly. In real arithmetic, the realized count is K_ℓ and $\sum_{i \in S_\ell} Y_i = D_\ell$. These are identities relative to generated budgets and counts, not constraints that force agreement with reference showers or incident energy.

3.4 Composite training objective

All components are optimized together with a weighted teacher-forced objective,

$$\begin{aligned} \mathcal{L} = & 2.574417\mathcal{L}_{\text{deposit}} + 0.160901\mathcal{L}_{\text{response}} + 2.159451\mathcal{L}_{\text{first}} + 0.536770\mathcal{L}_{\text{activity}} \\ & + 0.160901\mathcal{L}_{\text{profile}} + 0.160901\mathcal{L}_{\text{count}} + 1.324108\mathcal{L}_{\text{support}} + 1.477591\mathcal{L}_{\text{rank}} + 0.444960\mathcal{L}_{\text{share}}. \end{aligned} \quad (10)$$

The weights were frozen after train-only gradient-norm calibration on the shared condition encoder. If g_j is the median norm for component j and g_* their geometric mean, the unnormalized weight is $\text{clip}(g_*/g_j, 0.25, 4)$; final weights are rescaled to mean one. Deposit, activity, and support use binary cross entropies; first layer and counts use categorical cross entropies; profile and share use masked mean-squared flow-matching losses. Inactive count classes have weight 0.2. The support positive weight is the batch-wide valid negative-to-positive ratio clipped to $[1, 100]$. The rank term averages at most 256 random positive-negative comparisons per event, then averages over eligible events.

The selection objective expresses the response term in the transformed z measure. A density in deposited-energy coordinates would include the corresponding target-only Jacobian; it would still not be the likelihood of the clipped sampler. Figure 3 retains the historical objective as recorded. Its value is a selection criterion, not a calibrated measure of shower fidelity.

4 Checkpoint selection and validation methodology

4.1 Checkpoint selection

The model was optimized with AdamW, a 3×10^{-4} peak learning rate, segment-wise cosine annealing, batch size six, accumulation over four batches, gradient clipping at one, and no automatic mixed precision. The pilot lineage used training seed 20260723. A documented scheduler restart raised the learning rate at the epoch-70/71 boundary, so the history is not one uninterrupted cosine schedule. Figure 3 contains 104 recorded rows from absolute epochs 11–114. Epoch 90 has the lowest validation objective, 4.483768; the declared continuation at epochs 91–114 did not improve it. A separate continuation with an incorrect scheduler horizon was excluded from the accepted lineage. Earlier initialization stages and their complete

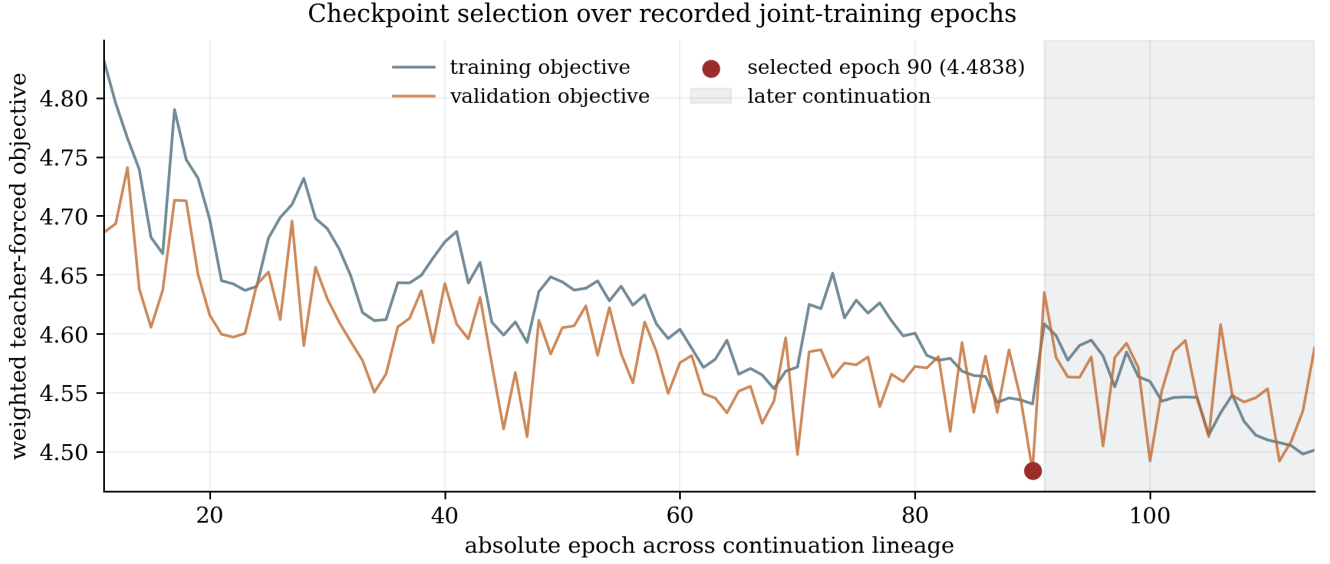


Figure 3: Training and validation objective over the accepted family. Epoch 90 is selected only by the minimum validation objective. The shaded epochs belong to the later declared continuation.

histories are not reconstructed by this extract. Our comparison concerns the selected single-seed artifact, not expected performance over retraining.

4.2 Metrics and denominators

Total-deposit means and standard deviations are computed in eight 25 GeV-wide incident-energy bins. Figure 4 reports

$$\Delta_\mu = 100 \frac{\mu_g - \mu_r}{\mu_r}, \quad \Delta_\sigma = 100 \frac{\sigma_g - \sigma_r}{\sigma_r}, \quad (11)$$

where g and r denote generated and reference events in the same bin. These are differences in empirical moments, not an incident-energy reconstruction resolution. The pooled Wasserstein-1 distance is accompanied by a reference-half comparison and a paired bootstrap that resamples conditions within energy bins. Those summaries describe finite-sample variation on this bank; they do not correct for repeated development use or supply fixed-condition ensembles.

A layer or channel is active when its stored energy is strictly positive; no detector threshold is applied. For a nonempty event with first and last active layers F and L , the occupied span is $L - F + 1$. We count inactive layers inside that span as $G = L - F + 1 - \sum_\ell A_\ell$. This counts missing layers, not contiguous inactive runs. An event has an interior gap when $G > 0$. These statistics exclude empty events. To avoid mixed denominators, Table 1 conditions every other activity and topology mean on nonempty events; conditional values are recovered from the released all-event means by dividing by the nonempty fraction. Weak components are connected components after discarding direction from the model graph. For a nonempty support S with weak components $C_1, \dots, C_m$, the largest-component fraction is $\max_j |C_j|/|S|$.

The only retained classifier number is a condition-only pipeline control with AUROC 0.500. A classifier receiving identical condition distributions in both classes has no shower information; this control cannot establish fidelity or rule out leakage in other features. Classifier two-sample tests require held-out discrimination [14]. For our paired design, reference/generated pairs must additionally remain together across partitions. Historical shower-aware classifier outputs are excluded because their split and feature construction do not support that comparison.

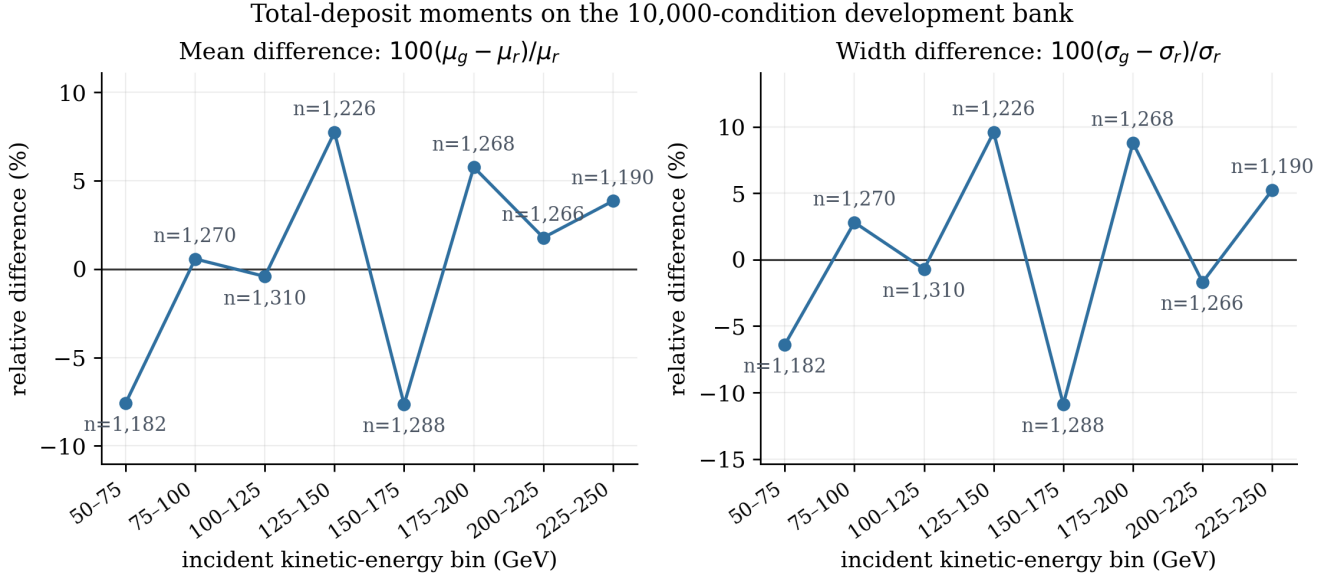


Figure 4: Differences in unconditional within-bin total-deposit moments. Every panel includes zero-deposit events. Labels give the number of matched conditions in each bin; no uncertainty bars are shown.

5 Validation results

5.1 Inclusive response and longitudinal energy allocation

The all-event mean total deposit is 4.369 GeV for the generator and 4.324 GeV for Geant4, a +1.0% difference. Across energy bins, mean differences range from -7.7% to $+7.7\%$ and standard-deviation differences range from -10.9% to $+9.6\%$ (Fig. 4). The pooled Wasserstein-1 distance is 0.0727 GeV. The approximate 2.5th and 97.5th percentile endpoints from 1,000 paired, energy-stratified bootstrap replicates are 0.0684 and 0.1984 GeV. These nominal 95% endpoints are exploratory; post-selection coverage was not established. A deterministic 5,000-versus-5,000 reference split gives 0.1475 GeV. Its smaller sample sizes differ from the 10,000-versus-10,000 model comparison, so it is not a calibrated noise floor or an equivalence threshold.

The total mean hides compensating subsystem errors. Mean ECAL energy is 2.275 GeV rather than 2.100 GeV (+8.3%), whereas mean HCAL energy is 2.094 GeV rather than 2.224 GeV (-5.8%). The mean longitudinal profile has relative L^1 discrepancy

$$\frac{\sum_{\ell=0}^{64} |\bar{D}_{\ell,g} - \bar{D}_{\ell,r}|}{\sum_{\ell=0}^{64} \bar{D}_{\ell,r}} = 0.071. \quad (12)$$

Figure 5 shows that the apparent total-energy agreement is partly cancellation rather than uniform layerwise agreement.

5.2 Longitudinal and readout-graph shower morphology

The clearest result appears after conditioning on nonempty events. The generator and reference have mean active-layer counts of 54.83 and 54.58, a 0.5% difference. Their arrangement is nevertheless different: 93.53% of generated events contain an interior gap compared with 52.43% of reference events, and the mean number of interior inactive layers is 6.05 rather than 2.10. The mean occupied span is also longer, 60.88 layers rather than 56.69. Thus similar active-layer means coexist with an extended span containing more inactive layers.

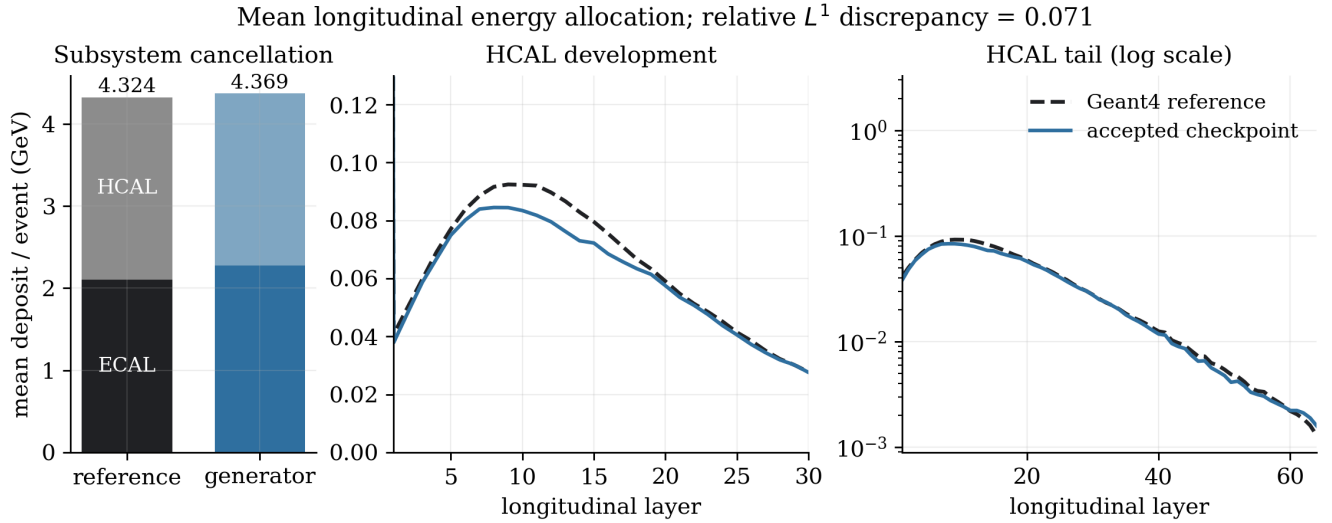


Figure 5: Mean longitudinal allocation on D_{dev} . The stacked bars expose opposite ECAL (lower) and HCAL (upper) shifts. The layer plots show HCAL layers only; layer 0 is isolated in the bars.

Table 1: Development-bank results. Except for the first row, quantities are conditional on nonempty events (9,907 reference and 9,858 generated). Values are descriptive because D_{dev} was repeatedly inspected.

Quantity	Geant4 reference	Generator	Gen./ref.
Zero-deposit events (%)	0.93	1.42	1.53
ECAL-start prevalence (%)	93.81	92.53	0.99
Mean first active layer	0.510	0.735	1.44
Mean active layers	54.58	54.83	1.005
Events with interior gaps (%)	52.43	93.53	1.78
Mean interior inactive layers	2.10	6.05	2.88
Mean active channels	1617.6	1588.8	0.982
Mean weak graph components	23.42	59.39	2.54
Mean largest-component fraction (%)	94.90	88.87	0.94

An analogous pattern appears at channel level. The mean number of active channels differs by -1.8% conditional on nonempty events, yet the generated support has 2.54 times as many weak readout-graph components. Its largest component contains 88.87% of active channels on average, compared with 94.90% in the reference. These statistics use the same graph that informs the model, so they are not independent geometric validation; they still show that exact cardinality realization does not imply a connected support law.

5.3 Numerical consistency of the decoder

The aggregate evaluator reports that all 1,250 batches of eight generated events passed its finite-value, nonnegativity, valid-support, count-realization, and budget checks. Maximum event and layer residuals were 1.14×10^{-5} and 3.05×10^{-5} GeV, respectively. For batch b , the implemented bound is

$$\epsilon_b = \max \left(2 \times 10^{-5} \text{ GeV}, 10^{-5} \max_{e \in b} |T_e| \right), \quad (13)$$

$$\max_{e \in b} \left| \sum_i Y_{ei} - T_e \right| \leq \epsilon_b, \quad \max_{e \in b, \ell} \left| \sum_{i \in \ell} Y_{ei} - D_{e\ell} \right| \leq \epsilon_b. \quad (14)$$

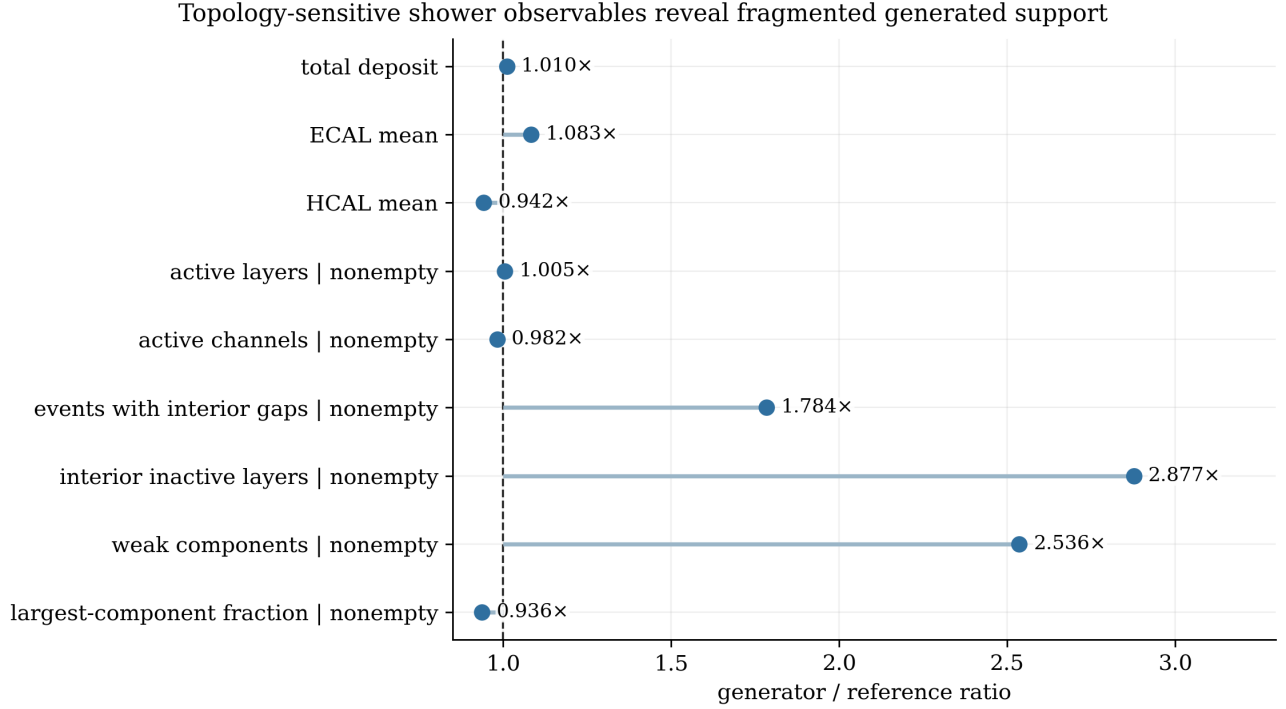


Figure 6: Generator-to-reference ratios. Total deposit and active counts lie near unity, while interior-inactivity and weak-component statistics do not. The graph quantities are conditional on nonempty events and use the readout graph defined in Sec. 2.

The same batch-wide tolerance is applied to every event and layer, rather than a relative bound evaluated at each individual budget. The largest effective bound was 4.94×10^{-4} GeV. This experiment uses the documented relative-tolerance policy introduced on 5 August 2026. The observed layer residual exceeds the earlier absolute-only 2×10^{-5} GeV rule; a pass under that older rule is not claimed. The aggregate preserves an all-batches-pass flag but not the individual batch reports, so its maxima alone cannot independently reconstruct every inequality.

Requested-versus-realized count mismatch, positive-support mismatch, invalid support, negative values, and nonfinite values were all zero. The raw-mode dust counter is identically zero because the detector threshold is zero; it is not a test for tiny positive deposits. These results establish the recorded numerical validity of this sample under the stated evaluator, not a universal floating-point guarantee.

6 Interpretation for surrogate validation

At the detector-observable level, the generator reproduces the mean active-layer and active-channel multiplicities much more closely than it reproduces the arrangement of those active elements. A 0.5% difference in mean active layers accompanies a factor-2.88 difference in interior inactive layers, while a -1.8% difference in mean active channels accompanies a factor-2.54 difference in weak graph components. The inclusive response shows the same pattern under coarser aggregation: a 1.0% difference in the total mean partly results from opposing ECAL and HCAL biases. The observed discrepancy is therefore a shower-morphology result, not a failure of the decoder to realize its sampled counts or energy budgets.

From the perspective of computational physics, the result separates three properties of a learned stochastic surrogate: numerical admissibility, agreement in low-order detector observables, and structural fidelity of the generated field. The first two do not imply the third. A minimal illustration uses four channels arranged along a path within one layer. The vectors $(1, 1, 0, 0)$ and $(1, 0, 1, 0)$ both have total energy two and two active channels, but their occupied subgraphs have one and two components. Point-

mass distributions on these vectors therefore have identical energy and count distributions and different connectivity. This algebraic example clarifies the logical distinction; the calorimeter comparison supplies the empirical scale.

The topology observables remain tied to the declared readout representation and a strict zero threshold. They do not isolate a pure dependence error at fixed one-channel marginals: changes in per-layer occupancy and energy mixing can also alter the measured fragmentation. The supported HEP statement is correspondingly concrete: agreement in inclusive response and occupancy multiplicity is insufficient to establish correct longitudinal and readout-graph shower morphology for this checkpoint.

The architecture offers hypotheses for the discrepancy. Independent Bernoulli activity restricts residual dependence after conditioning on total deposit and first layer; graph scores followed by top- k sampling do not impose connectivity; and reference-conditioned training differs from ancestral generation. The present comparison does not isolate those mechanisms. An oracle cascade, substituting reference quantities at one stage at a time, would test their contributions. A controlled sequential-activity model and a structured-support model would then test possible repairs. More training data is also an unresolved factor because this checkpoint used only the pilot.

CaloChallenge and CaloDREAM already combine shower-shape diagnostics with multivariate evaluation [6, 7]. Full-physics studies extend this to reconstruction [8]. The present result complements that work with interpretable gaps and readout-graph components; it does not establish that existing evaluation methods systematically miss such failures.

7 Scope and requirements for physics validation

This study establishes a detector-specific validation result for one selected pilot-trained checkpoint, one fixed readout, and one repeatedly inspected development bank. It does not establish overall physics fidelity, architecture superiority, or acceleration. Agreement in the reported means is not agreement of the corresponding full distributions. The support diagnostics are threshold- and graph-dependent: ganged-channel centroids lose physical detail, and both generation and evaluation use the same adjacency. Missing per-event arrays prevent a new uncertainty analysis of the headline topology ratios; those ratios are descriptive observations rather than population estimates with calibrated confidence intervals.

A broader claim about model performance would require independent training seeds, a locked evaluation bank with documented event identities, and energy- and angle-resolved comparisons. Repeated reference and generated showers at fixed conditions would assess conditional variability. Cap activation, reference cap exceedance, CLR clipping, solver convergence, positive-threshold sweeps, and an independent physical-neighbor graph would test sampler and representation sensitivity. A matched baseline suite and an oracle cascade would distinguish limited pilot training from architectural restrictions.

For application as a detector surrogate, further work must establish reconstructed-observable agreement, memorization controls, and end-to-end timing including solver, decode, transfer, and serialization costs. The released evidence also lacks a complete Geant4 version, physics list, production-cut, material, field, and seed specification. These limitations bound extrapolation beyond the measured comparison; they do not alter the reported discrepancy between occupancy multiplicity and shower morphology on D_{dev} .

8 Conclusion

We have presented a topology-sensitive validation study of a constraint-preserving generative surrogate for single-neutron showers in a 6,790-channel Zero Degree Calorimeter readout. For the analyzed checkpoint, the generated events satisfy the recorded decoder-level numerical checks, and the mean active-layer and active-channel multiplicities differ from the Geant4 reference by 0.5% and -1.8% . Nevertheless, interior inactive layers and weak readout-graph components increase by factors of 2.88 and 2.54. The percent-level agreement in mean total deposit also contains compensating ECAL and HCAL biases.

The detector-level conclusion is that inclusive response and occupancy agreement do not establish correct shower morphology. The computational conclusion is that constraint preservation guarantees admissibility with respect to the decoder, not fidelity to the reference stochastic process. Validation of learned calorimeter surrogates should therefore report numerical consistency, inclusive detector observables, and correlation- or topology-sensitive shower observables as distinct tests. The present result supports that requirement for the specified checkpoint and development bank; broader performance claims require the independent seeds, locked evaluation, reconstruction studies, and timing measurements described above.

Code and data availability

The accompanying build package contains the manuscript source, aggregate validation report, event-independent geometry, training-history extract, figure builder, and QA records. The project repository is <https://github.com/JulianAttemptsCoding/fast-mc-paper>. The accepted checkpoint is identified by SHA-256 491284c7423f365230d34b0443f95aa4888ec770bdc673c4c979897bad8acbce; the frozen runtime configuration is identified by SHA-256 116bc8c220b07ce54ae07196bdd6ed8e835775c8c937182a209a799dc94ae9c5. The collaboration-owned event file and checkpoint are not redistributed. Public aggregates reproduce the figures and quoted battery statistics, but they cannot retrain the model or reconstruct event-level tests.

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Author contributions

Julian Juan: conceptualization, methodology, software, investigation, visualization, and writing.

Use of generative AI

Generative-AI tools were used for editorial assistance, code generation, literature discovery, and adversarial manuscript auditing. Numerical statements were checked against versioned project artifacts during this revision; the author retains responsibility for verifying and approving the final manuscript.

A Artifact identity and audit boundary

The aggregate battery report has schema version 3 and records 10,000 generated and 10,000 validation-reference events, zero test events, generator seed 20260723, evaluator seeds 20260804–20260806, and checkpoint epoch 90. The checkpoint was selected by the lowest validation objective over its declared horizon. The report’s checkpoint hash matches its provenance sidecar. A queued copy has a different file hash because its serialized container differs; no metric is joined by file hash across those two containers in this paper.

The package includes source-evidence extracts binding the pilot training count to the frozen configuration, the canonical preparation counts, calibration weights, and evaluator contract. The training split and diagnostic split have different hashes because they serve different roles. The paper rebuilds its figures from aggregates; it does not independently reproduce training, the original configuration, or per-event

evaluation. Archived QA records from earlier manuscript versions are historical and do not validate this revision.

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